

Monetary–Fiscal Coordination as a State-Contingent Dynamic Game: Steady States, Shocks, Commitment, and an Australian Calibration

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Abstract

This paper develops a tractable New Keynesian game between an independent central bank and a fiscal authority, calibrated to Australian data. The model keeps the structural economy and a single social-welfare criterion fixed while allowing the two agencies to place different relative weights on inflation and the output gap. We first solve cooperative, simultaneous Nash, and Stackelberg versions of a finite-horizon linear–quadratic game. With no shock and zero gap targets, every solution has the same zero steady state: strategic interaction is present in response derivatives but dormant in realised policy. Shocks activate it. A demand contraction principally creates a burden-sharing and free-riding problem; a cost-push shock creates an inflation–output conflict and mutually reinforcing instrument responses, with monetary tightening and fiscal expansion offsetting one another in aggregate demand. At the baseline calibration, non-cooperation raises true social loss by 94.8 per cent after a demand shock and 106.3 per cent after a cost-push shock. These calculations clarify both the value and the limitation of the current open-loop approach. It is a useful full-path-commitment benchmark, but it is not the appropriate terminal solution concept for the original draft, in which lagged policy, adjustment costs, future shocks, and endogenous commitment technology are central. The natural main equilibrium is instead a stochastic feedback (Markov-perfect) Nash game, supplemented by feedback Stackelberg and an ex ante commitment-technology game. We state that model explicitly and describe the next computational experiment.

Keywords: monetary–fiscal coordination; dynamic games; Markov-perfect equilibrium; strategic complementarity; fiscal dominance; Australia.
JEL codes: E52, E58, E61, E62, C73.

1 Introduction

Monetary and fiscal authorities act on the same economy but need not evaluate the resulting inflation, activity, debt, or instrument movements in the same way. This observation is enough to create a policy game even when both institutions are benevolent and the model contains no political distortion. It does not, however, imply that every shock produces the same kind of game. Whether the instruments are strategic complements or substitutes, whether an authority free rides, and whether leadership improves outcomes depend on the shock, nominal rigidity, the state variables, and the timing of commitment.

The motivating draft distinguishes two mechanisms. Following a demand shock, both authorities want to stabilise activity but may try to shift the adjustment cost to the other

authority—a “GFC effect.” Following a supply disturbance, the central bank’s concern for inflation can produce tightening while the fiscal authority’s concern for activity produces expansion. The policies then offset one another at the macroeconomic level while becoming mutually reinforcing in their numerical instrument settings—a “post-COVID effect.” The draft also makes prior policy a state through adjustment costs and asks whether institutions deliberately acquire commitment technology. These are recursive, state-contingent questions.

The existing computational model answers a narrower but useful question. At the beginning of a finite horizon, after a shock path is known, the authorities choose complete instrument paths. This is an open-loop game. It isolates the policy-mix externality, permits clean welfare comparisons, and provides a benchmark against which a recursive solution can be checked. It cannot, by itself, answer how authorities reoptimise when new information arrives or how today’s instrument choice changes tomorrow’s strategic state.

This revision makes that boundary explicit. It contributes four results. First, it embeds fiscal policy and debt in a transparent extension of the three-equation New Keynesian model. Second, it separates agency mandates from true social welfare. Third, it establishes experimentally that the calibrated no-shock steady state contains no realised conflict, while shocks generate large policy-game losses. Fourth, it formulates the stochastic Markov-perfect game that corresponds to the original draft and explains why open-loop Nash and open-loop leadership should be retained as benchmarks, not treated as the final equilibrium concepts.

2 From three equations to a monetary–fiscal game

2.1 The private-economy block

Let x_t denote the output gap, π_t inflation relative to target, i_t the monetary-policy instrument relative to neutral, and g_t a fiscal-demand instrument, where $g_t > 0$ is expansionary. A compact forward-looking model is

$$x_t = \mathbb{E}_t x_{t+1} - \sigma(i_t - \mathbb{E}_t \pi_{t+1}) + \psi g_t + d_t, \quad (1)$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t, \quad (2)$$

$$b_{t+1} = \rho_b b_t + \chi_g g_t + \chi_i i_t - \chi_\pi \pi_t. \quad (3)$$

Here d_t is a demand disturbance, u_t is a cost-push disturbance, and b_t is the public-debt gap. The signs match the code: a larger i_t is tighter and a larger g_t is more expansionary. Equation (1) is the IS curve, (2) is the New Keynesian Phillips curve, and (3) is a deliberately parsimonious debt law.

It is potentially misleading to call this merely a “four-equation model.” The conventional three-equation model consists of IS, Phillips, and an exogenous monetary rule. Once both policies are optimised, the Taylor rule is replaced by the central bank’s first-order condition and a fiscal first-order condition is added. With debt retained, the equilibrium has five blocks: IS, Phillips, debt accumulation, monetary optimality, and fiscal optimality. For active–passive regime comparisons we also use the explicit rule closure

$$i_t = \rho_i i_{t-1} + (1 - \rho_i)(\phi_\pi \pi_t + \phi_x x_t) + \varepsilon_t^i, \quad (4)$$

$$g_t = \rho_g g_{t-1} - (1 - \rho_g)(\gamma_b b_t + \gamma_x x_t) + \varepsilon_t^g. \quad (5)$$

Equations (4)–(5) are an alternative closure, not additional behavioural assumptions inside the optimising game.

2.2 One social objective, two agency mandates

The primitive social loss is fixed across every game:

$$\mathcal{L}^S = \frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\omega_{\pi}^S \pi_t^2 + \omega_x^S x_t^2 + \omega_b^S b_t^2 + \omega_i^S i_t^2 + \omega_g^S g_t^2 + \omega_{\Delta i}^S (i_t - i_{t-1})^2 + \omega_{\Delta g}^S (g_t - g_{t-1})^2]. \quad (6)$$

It is the welfare yardstick and therefore does not change when the strategic regime changes. Cooperation means that a single planner minimises (6); it does not mean adding together two differently weighted agency losses.

The agency losses have the same form but different inflation and output-gap weights:

$$\mathcal{L}^j = \frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\omega_{\pi}^j \pi_t^2 + \omega_x^j x_t^2 + \omega_b b_t^2 + \omega_i i_t^2 + \omega_g g_t^2 + \omega_{\Delta i} (\Delta i_t)^2 + \omega_{\Delta g} (\Delta g_t)^2], \quad (7)$$

where $j \in \{M, F\}$. In the baseline calibration the monetary authority is more inflation averse, $(\omega_{\pi}^M, \omega_x^M) = (1.5, 0.25)$, while the fiscal authority is more output oriented, $(\omega_{\pi}^F, \omega_x^F) = (0.5, 0.75)$. All other weights are held fixed. The social weights are $(\omega_{\pi}^S, \omega_x^S) = (1, 0.5)$. Thus the experiment changes mandates, not technology or the definition of welfare.

This distinction also explains why cooperation can involve a larger response by one instrument than a decentralised equilibrium. Cooperation internalises both instruments' effects and assigns the policy mix according to (6). Nash policymakers instead minimise different losses and may partially offset or free ride on one another. A larger cooperative movement is not generated by changing the planner's weights across cases; it is an equilibrium allocation of a fixed social objective.

3 Timing, states, and equilibrium concepts

3.1 Shocks and the strategic state

For the intended stochastic model, take

$$d_{t+1} = \rho_d d_t + \varepsilon_{t+1}^d, \quad u_{t+1} = \rho_u u_t + \varepsilon_{t+1}^u, \quad (8)$$

with innovations not known at t . A minimal predetermined state is

$$z_t = (b_t, i_{t-1}, g_{t-1}, d_t, u_t)'. \quad (9)$$

Lagged instruments matter because adjustment costs make past policy affect the current marginal cost of action. Debt transmits fiscal and monetary choices across time. Persistent shocks change the distribution of future states. If private expectations are solved under commitment, promised inflation or the planner's multipliers must also be included in an augmented state.

The timeline is: the state z_t is observed; private agents form rational expectations using anticipated policy functions; the authorities choose their instruments according to the equilibrium timing; outcomes occur; and new innovations produce z_{t+1} . This timing makes the policies contingent on the realised state, not just on calendar time.

3.2 Feedback Nash and Markov perfection

Let the strategy functions be

$$i_t = F_M(z_t), \quad g_t = F_F(z_t). \quad (10)$$

A stationary Markov-perfect Nash equilibrium is a pair (F_M, F_F) such that, given the other authority's policy function and rational private expectations, each authority solves its Bellman problem at every state. Schematically,

$$V_M(z) = \min_i \{ \ell_M(z, i, F_F(z)) + \beta \mathbb{E}[V_M(z') \mid z, i, F_F(z)] \}, \quad (11)$$

$$V_F(z) = \min_g \{ \ell_F(z, F_M(z), g) + \beta \mathbb{E}[V_F(z') \mid z, F_M(z), g] \}. \quad (12)$$

In the linear-quadratic approximation, value functions are quadratic and the policy functions are linear, $i_t = -K_M z_t$ and $g_t = -K_F z_t$. The equilibrium is obtained from coupled Riccati equations, with stabilising and determinacy conditions imposed on the joint private-policy system.

This is the natural main solution concept for the original draft. The authority that deviates today understands how the deviation changes debt and the next period's adjustment-cost state, but it cannot bind its future self to ignore information that will later be available. Consequently the solution speaks directly to credibility, reoptimisation, and policy used as a strategic state.

3.3 What open-loop equilibrium does and does not mean

The current numerical game instead selects complete paths $I = (i_0, \dots, i_{T-1})'$ and $G = (g_0, \dots, g_{T-1})'$ at date zero. Conditional on a known shock path, the linear economy can be written

$$q = q^0 + A_I I + A_G G, \quad (13)$$

where q stacks macroeconomic outcomes. Quadratic agency losses imply affine best responses

$$I = a_M + B_M G, \quad G = a_F + B_F I. \quad (14)$$

The open-loop Nash path solves the two equations simultaneously. The round-trip matrix $B_M B_F$ measures the strength of iterative reaction, and $\rho(B_M B_F) < 1$ is a useful local convergence diagnostic.

Open-loop equilibrium is appropriate if the economic experiment is explicitly one of full path commitment after the shock is observed and no unanticipated innovation subsequently arrives. It is also valuable as a transparent benchmark. It is not sufficient when the object of interest is policy under recurring shocks, discretion, or strategic investment in adjustment costs. The path is time-indexed rather than state-contingent, and an authority does not reoptimise after histories that were not anticipated at date zero.

Nor does the spectral radius alone identify strategic complementarity. It is unsigned and summarises a high-dimensional round trip. Classification requires the signed cross-derivatives $\partial i_t / \partial g_s$ and $\partial g_t / \partial i_s$, evaluated at an economically specified state or shock. With the present sign convention, positive contemporaneous slopes are complementarity in instrument levels: fiscal expansion elicits monetary tightening, and tighter money elicits more fiscal expansion. The two policies then offset one another in aggregate demand.

3.4 Leadership and the commitment meta-game

The existing Stackelberg calculations are also open loop. A leader chooses a complete path while internalising the follower’s complete-path best response. This can make monetary and fiscal leadership look similar because both leaders have nearly the same ability to exploit the same linear reaction mapping and because the calibration is close to a common welfare problem once the follower is internalised.

The recursive counterpart is feedback Stackelberg equilibrium. There the leader anticipates the follower’s state-contingent response at every future state. The original draft additionally suggests a prior institutional stage: before shocks are realised, authorities choose commitment or adjustment-cost parameters, say λ_M and λ_F ; after shocks, they play the feedback policy game. Formally, the ex ante problem is

$$\lambda_j^* \in \arg \min_{\lambda_j} \mathbb{E}[V_j(z_0; \lambda_j, \lambda_{-j})], \quad (15)$$

where V_j is generated by the continuation MPE. This separates leadership within the stabilisation game from the acquisition of commitment technology.

4 The steady state and the shock-contingent type of game

4.1 No-shock steady state

Set $d = u = 0$ and use zero gap targets. Equations (1)–(3) and either policy closure admit

$$x^* = \pi^* = b^* = i^* = g^* = 0. \quad (16)$$

Every term in each quadratic loss is then zero. Because policy costs are strictly convex, no authority can improve on this allocation by moving its instrument. The numerical zero-shock experiment verifies that cooperation, Nash, monetary leadership, and fiscal leadership all return exactly this steady state.

This finding should be stated carefully. It does not say that institutions or coordination are irrelevant in normal times. Reaction slopes and the welfare consequences of future states remain latent, and an ex ante commitment choice can still matter because shocks occur with positive probability. It says that the present aligned-target calibration does not generate a realised steady-state policy conflict. A non-zero steady-state game would require an additional wedge such as different inflation targets, a fiscal spending bias, distortionary tax objectives, a debt target conflict, or seigniorage pressure.

Table 1: Numerical no-shock steady state

Strategy	social loss	$\max x $	$\max \pi $	$\max i $	$\max g $
Cooperation	0	0	0	0	0
Simultaneous Nash	0	0	0	0	0
Monetary leader	0	0	0	0	0
Fiscal leader	0	0	0	0	0

4.2 A taxonomy by shock

The shock determines which disagreement is economically active. Table 2 gives the disciplined interpretation intended for the full feedback model.

Table 2: Shock-contingent strategic interaction

State	Main policy trade-off	Likely strategic mechanism	Empirical diagnostic
No shock, aligned targets	None realised at the zero-gap steady state	Dormant reaction structure; coordination matters ex ante	Identical zero policies and losses
Demand contraction	Both agencies favour expansion, but dislike changing their own instruments	Burden sharing and free riding; stabilisation efforts can be substitutes	Signed cross-responses and each authority’s contribution to closing the output gap
Cost-push inflation	Inflation stabilisation conflicts with output stabilisation	Monetary tightening and fiscal expansion can be complements in instrument levels but offsets in demand	Signed policy slopes, policy-mix distance, inflation/output outcomes
Debt or solvency shock	Price stability, debt stabilisation, and valuation effects interact	Active–passive regimes and possible fiscal dominance	Determinacy, debt backing, and nominal-liability valuation

Nominal rigidity is pivotal after a cost-push shock. When κ is finite, lower inflation requires an output sacrifice. Because the monetary authority places greater weight on inflation and the fiscal authority places greater weight on output, their preferred points on that trade-off differ. The fiscal response then raises the central bank’s desired rate, while the monetary response raises the fiscal authority’s desired support. This is precisely the case in which feedback, expectations, and commitment may alter the result most.

5 Australian calibration and computation

The code is written entirely in Julia. The descriptive IS and Phillips coefficients are estimated using quarterly Australian data over 1993Q1–2019Q4, with public series cached alongside checksums for reproducibility. The sample ends before the pandemic so that the cost-push experiment is not mechanically fit to the episode it is intended to illuminate. The debt block and welfare weights are transparent calibrations rather than claimed structural estimates.

Table 3: Selected baseline parameters

Parameter	Value	Interpretation
β	0.9900	quarterly discount factor
σ	0.03695	semi-elasticity to the real-rate gap
ψ	0.05142	public-demand multiplier in the IS equation
κ	0.15511	Phillips-curve slope
ρ_b	0.995	debt persistence
$(\omega_\pi^S, \omega_x^S, \omega_b^S)$	(1, 0.5, 0.15)	social weights
$(\omega_\pi^M, \omega_x^M)$	(1.5, 0.25)	monetary mandate
$(\omega_\pi^F, \omega_x^F)$	(0.5, 0.75)	fiscal mandate

For each deterministic shock path, the program constructs the linear path map (13), derives exact quadratic best responses, and solves four regimes: the social planner, simultaneous Nash, monetary leadership, and fiscal leadership. It also varies mandate divergence and

policy adjustment costs, and solves the explicit rule system in monetary- and fiscal-dominance analogues.

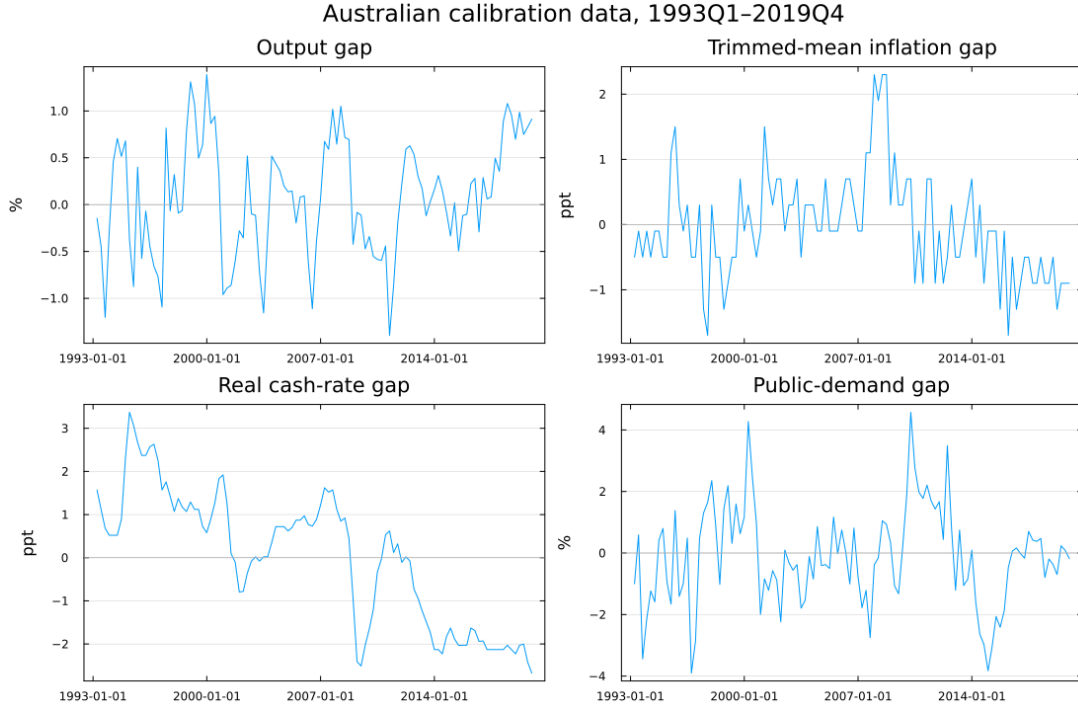


Figure 1: Australian data used for the descriptive calibration.

6 Results

6.1 Shocks activate the coordination problem

Figure 2 scales the demand and cost-push innovations from zero to 1.5 sample standard deviations. At zero, the Nash welfare gap and the distance between cooperative and Nash policy paths are both zero. As shocks grow, the path distance rises linearly and excess Nash loss rises quadratically, as expected in a linear–quadratic model. The supply-shock curves are steeper: the shock activates the disagreement over the inflation–output trade-off.

6.2 Demand contraction: burden sharing and free riding

After a one-standard-deviation contractionary demand shock, cooperative social loss is 0.0972 and simultaneous Nash loss is 0.1892, a 94.8 per cent increase. The peak monetary response is 0.067 under cooperation but 0.306 under Nash; the peak fiscal response is 0.075 under cooperation and 0.257 under Nash. The large gross movements do not indicate stronger stabilisation. They partly reflect the authorities reacting to and offsetting the policy mix generated by the other authority. The output peaks remain similar, while debt accumulation is much larger under Nash.

The open-loop leadership outcomes are close to the planner: social loss is 0.0991 with monetary leadership and 0.0986 with fiscal leadership. This result shows that internalising the follower’s reaction removes most of the static path externality. It should not yet be read as evidence that the institutional identity of the leader is unimportant. Feedback leadership may

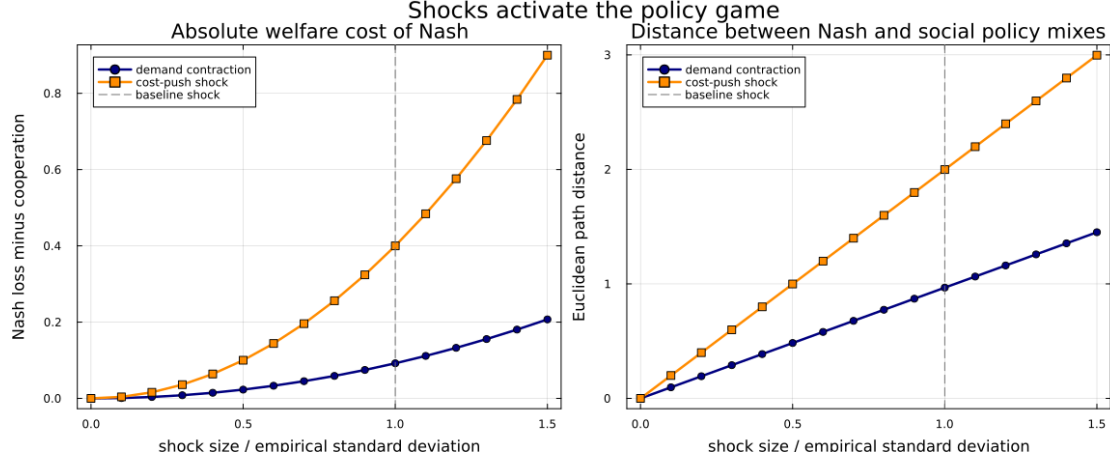


Figure 2: Shock activation of the coordination problem. The left panel is $\mathcal{L}_N^S - \mathcal{L}_C^S$; the right panel is the Euclidean distance between the Nash and cooperative instrument paths.

differ because debt and lagged instruments change subsequent bargaining positions, while an ex ante commitment choice changes continuation values before the shock occurs.

6.3 Cost-push shock: the offsetting-policy game

After a one-standard-deviation cost-push shock, cooperative social loss is 0.3762 and Nash loss is 0.7762, an increase of 106.3 per cent. Nash produces a peak monetary response of 0.607 and a peak fiscal response of 0.500, compared with 0.050 and 0.046 under cooperation. This is the clearest baseline example of strategic complementarity in instrument levels. The central bank tightens against inflation; the fiscal authority supports activity; each action raises the other's incentive to respond. Because i_t and g_t enter the IS curve with opposite signs, complementarity between the numerical instruments means macroeconomic offset, not common-direction stimulus.

Again, open-loop leadership nearly restores cooperative welfare: social loss is about 0.386 under either leader. The similarity follows from the common linear path technology and the leader's internalisation of the follower's full-path response. Whether it survives recursive reoptimisation is a substantive question for the feedback Stackelberg experiment.

6.4 Mandate divergence and the coordination boundary

At the baseline mandate divergence, the monetary best-response norm is 1.313 and the fiscal norm is 0.742. The round-trip spectral radius is 0.950, close to the unit boundary. This explains why iterated reactions can amplify small differences and why decentralised policies are so much larger than cooperative policies. It does not establish that every element of both reaction matrices is positive, and it should not be used as a shock-invariant label for the game.

Figures 5 and 6 show that stronger mandate divergence and lower instrument costs move the open-loop system toward the coordination boundary. The economically important next step is to reproduce these maps using the eigenvalues of the closed-loop stochastic system and signed, state-contingent cross-responses.

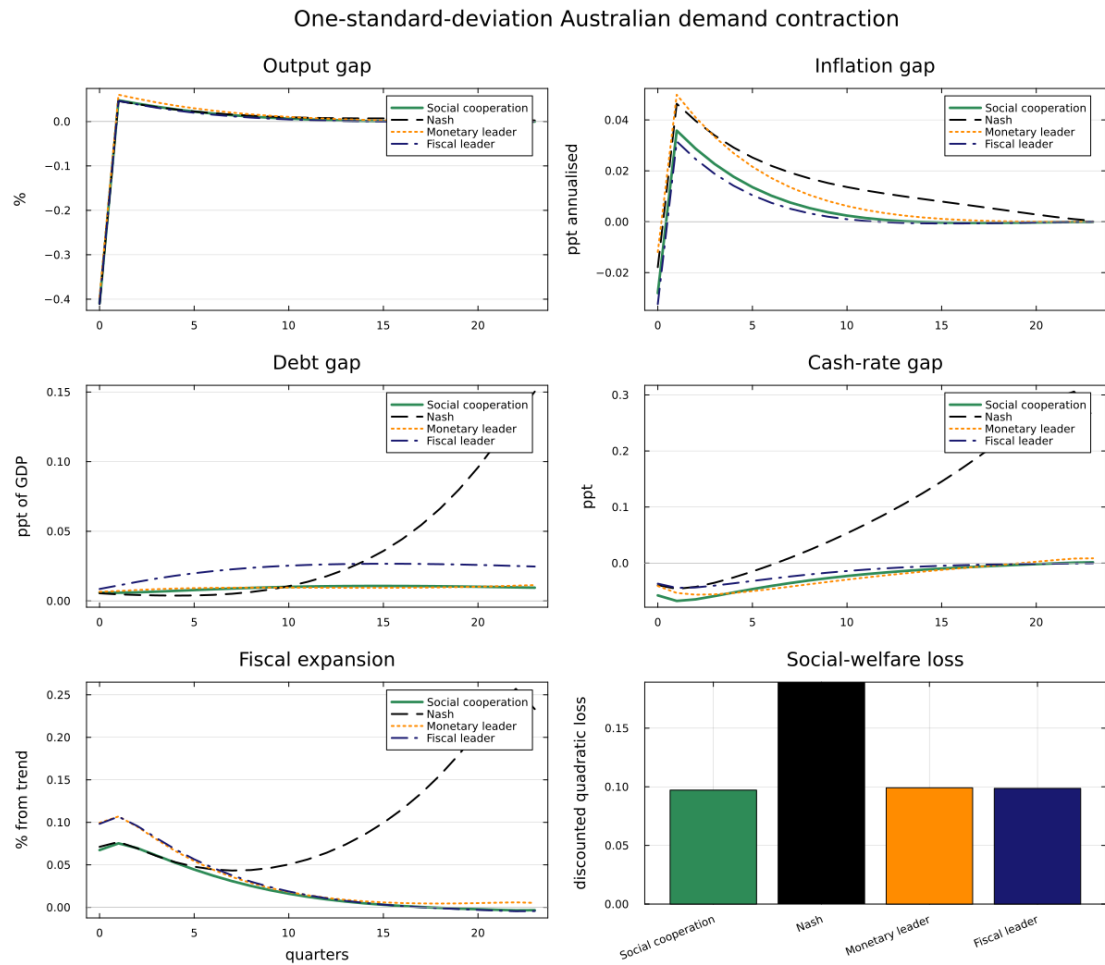


Figure 3: Responses to a contractionary demand shock under cooperation, simultaneous open-loop Nash, and open-loop leadership.

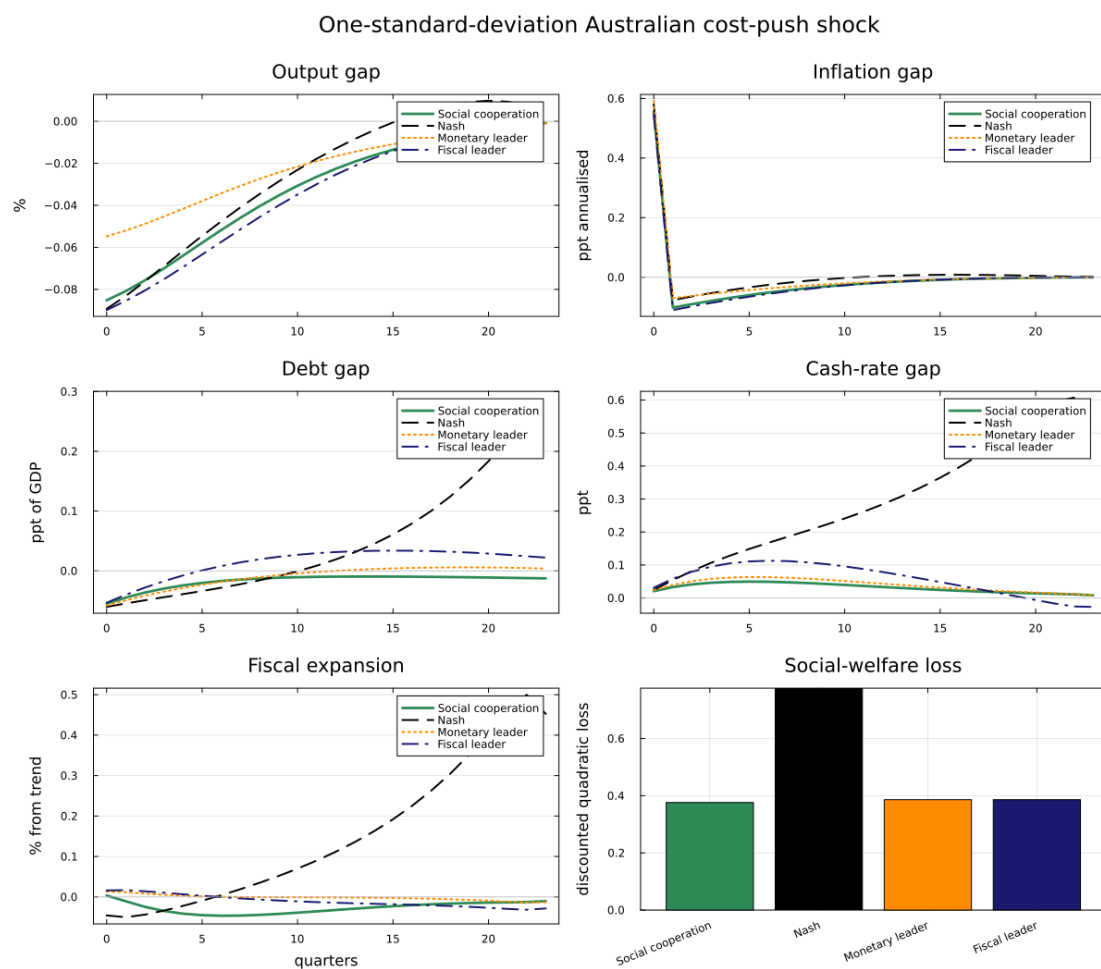


Figure 4: Responses to a cost-push inflation shock.

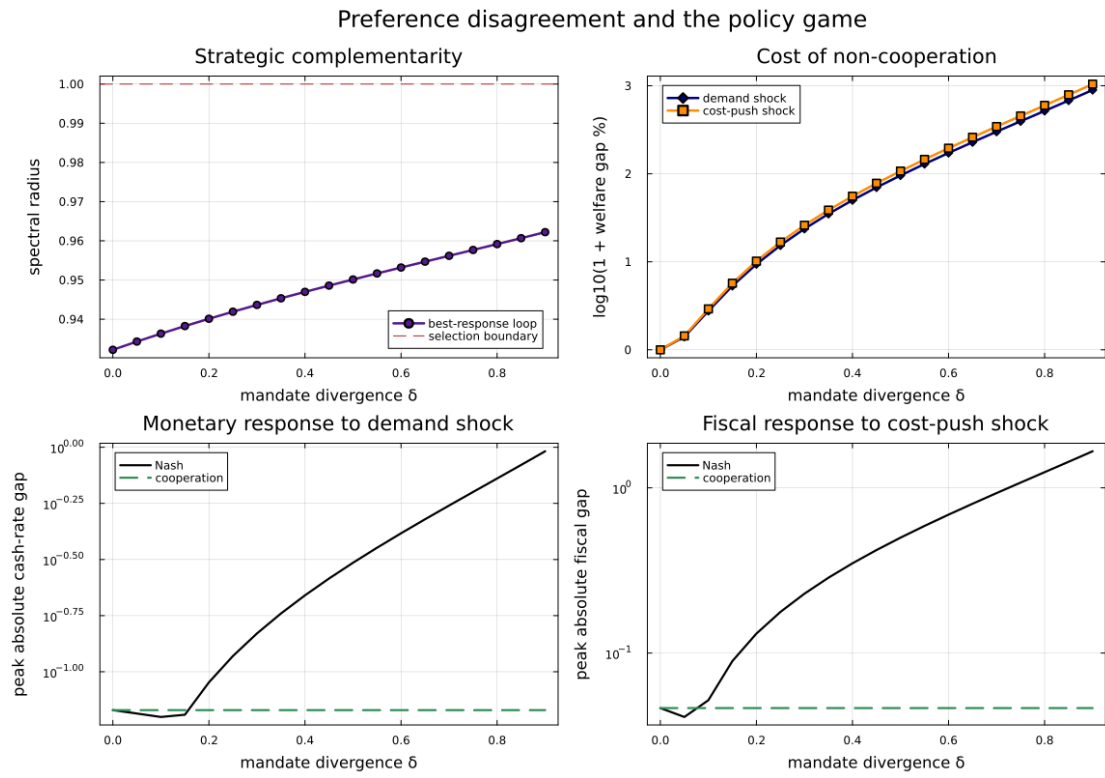


Figure 5: Open-loop reaction strength and welfare as agency mandates diverge.

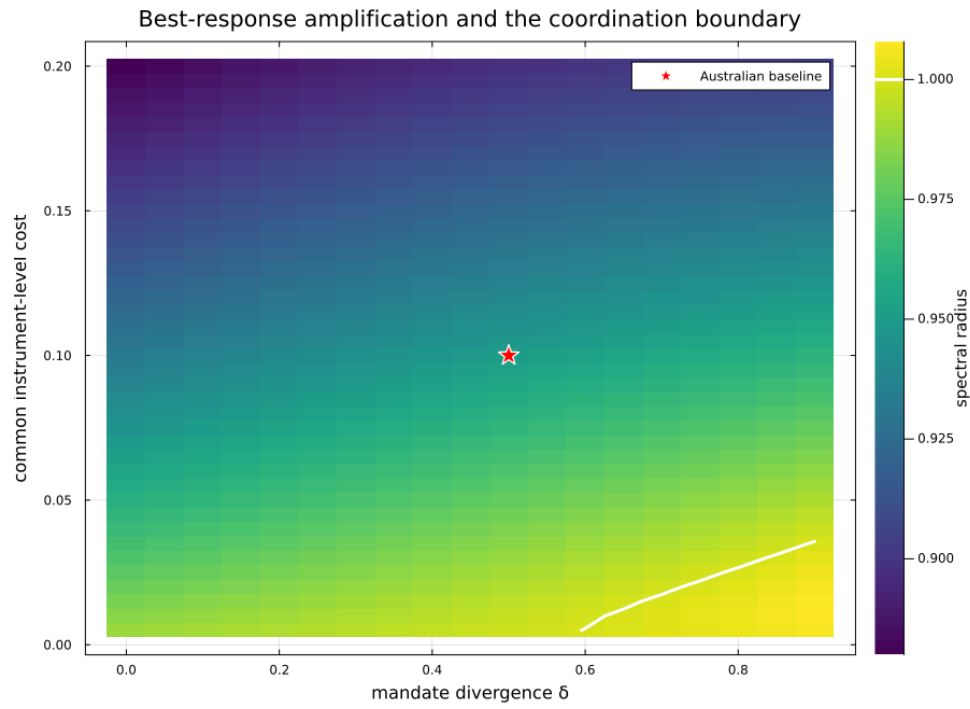


Figure 6: The open-loop coordination boundary across mandate divergence and policy-cost calibrations.

6.5 Does the model nest fiscal dominance?

The rule closure nests an active–passive regime comparison in the sense of [Leeper \(1991\)](#). In the monetary-dominance configuration, the interest-rate rule responds strongly to inflation and fiscal policy stabilises debt. In the fiscal-dominance analogue, monetary policy responds weakly and fiscal policy is less debt responsive. The latter produces peak inflation of 0.757 versus 0.404 and a peak policy rate of 0.244 versus 0.480 in the monetary-dominance case. Figure 7 displays the paths.

This is a nesting of the active–passive logic, not a complete fiscal theory of the price level. The parsimonious debt equation lets inflation reduce the real debt gap, which can make the calibrated quadratic social loss lower in the fiscal-dominance analogue. A full fiscal-dominance claim requires nominal government liabilities, a consolidated intertemporal budget constraint, explicit primary-surplus backing, and equilibrium price-level valuation. The current result should therefore be read as a regime diagnostic rather than a welfare endorsement.

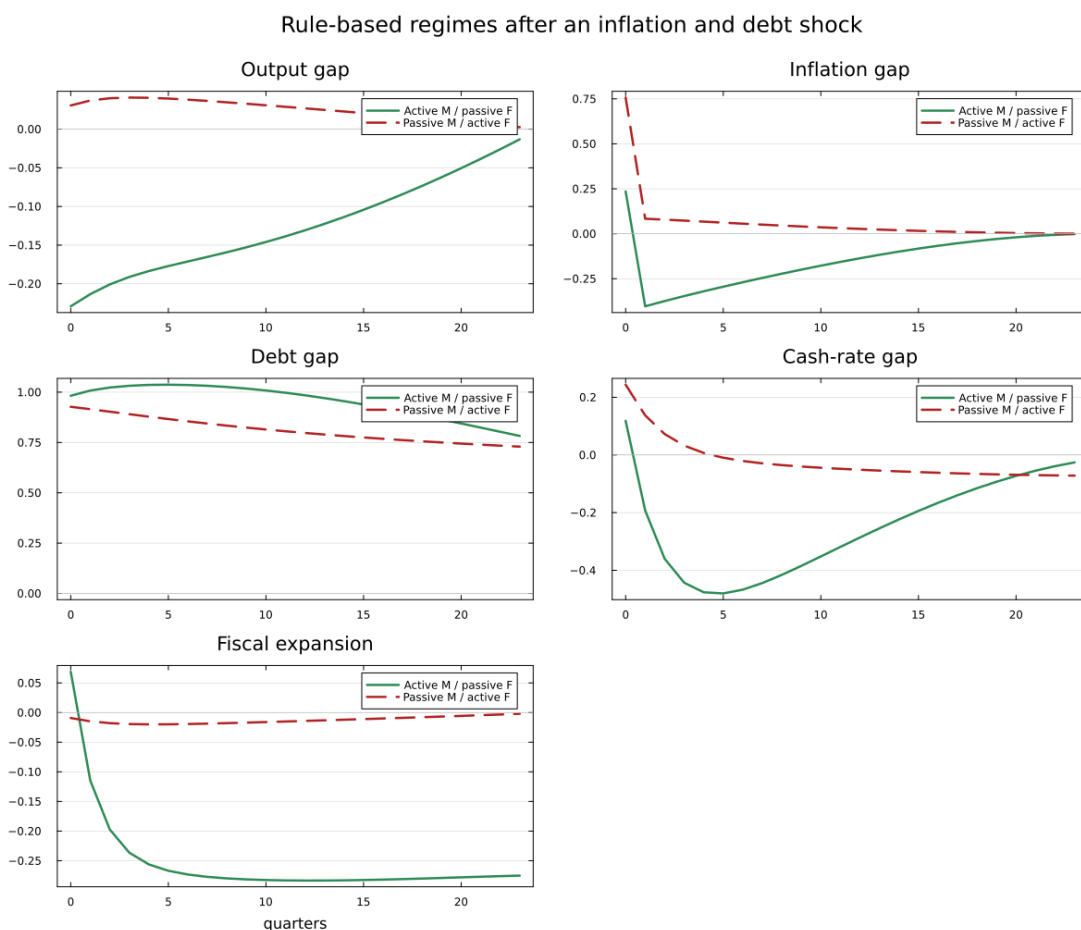


Figure 7: Active–passive rule-system comparison.

7 Is the current approach sufficient?

The answer is no if the objective is to implement the full economic mechanism in the motivating draft. It is sufficient for a first, controlled benchmark: it demonstrates that mandate differences alone can create large policy-mix externalities, holds social primitives fixed, and

distinguishes the no-shock steady state from shock responses. But three omissions matter for the central claim.

First, a known deterministic shock path eliminates the arrival of new information. Second, open-loop paths prevent policies from responding to off-equilibrium states, whereas debt and lagged instruments are precisely the states through which strategic commitment operates. Third, open-loop leadership conflates leadership with full path commitment; it does not reveal how a leader manipulates the follower’s future state-contingent rule. Nominal rigidity makes these omissions especially material after a cost-push shock, because expected future policy affects current inflation and output.

Table 4: Role of the alternative solution concepts

Concept	What it identifies	Limitation or role
Open-loop Nash	Policy-mix externality when complete paths are chosen after a known shock	Useful commitment benchmark; no reoptimisation after new states
Open-loop Stackelberg	Value of internalising the follower’s complete-path reaction	Leadership and full-path commitment are bundled together
Feedback/MPE Nash	Discretionary, state-contingent interaction under recurring shocks	Appropriate main equilibrium for debt and lagged-policy states
Feedback Stackelberg	State-contingent strategic leadership	Required to test whether the two leadership regimes genuinely differ
Commitment meta-game	Ex ante choice of adjustment costs, delegation, or rules	Implements the draft’s endogenous commitment/prisoner’s-dilemma mechanism

8 Proposed stochastic experiment

The next version should preserve the same structural coefficients, mandate weights, and true social loss, so that differences come only from equilibrium timing. The following sequence directly tests the draft’s mechanism.

1. Estimate or transparently calibrate (ρ_d, ρ_u) and innovation covariance in (8). Retain b_t, i_{t-1}, g_{t-1} as predetermined states.
2. Solve the cooperative stochastic regulator and the stabilising feedback Nash equilibrium from the coupled Riccati system. Verify determinacy, closed-loop stability, and uniqueness; where multiple stabilising solutions exist, report the selection criterion.
3. Report signed state-contingent response coefficients, particularly $\partial F_M / \partial u$, $\partial F_F / \partial u$, and the cross-policy effects implied by each authority’s first-order condition. Do not classify the game using only an unsigned spectral radius.
4. Solve monetary- and fiscal-leader feedback Stackelberg equilibria. Compare their continuation values and policy functions with open-loop leadership to isolate the value of state-contingent commitment.
5. Before the shock process begins, allow authorities to choose adjustment cost or delegation parameters. Solve (15) and compare the non-cooperative institutional choice with

the socially preferred one. This is the direct test of strategic investment in commitment technology.

6. Repeat the demand, cost-push, and debt experiments. For the demand shock, measure burden sharing and free riding. For the cost-push shock, measure offsetting policy, instrument complementarity, and excess variance. For debt stress, distinguish policy coordination from active-passive determinacy and fiscal valuation.

This design lets the steady-state and shock insights reinforce rather than contradict each other. The zero-gap deterministic steady state is a common reference point. Institutions matter because they determine the feedback rules that will be used when shocks move the economy away from that point. Demand and cost-push shocks then activate different strategic margins, while the ex ante commitment stage determines whether authorities choose institutions that mitigate or intensify those margins.

9 Conclusion

Different relative inflation and output weights are sufficient to generate a substantive monetary-fiscal game once shocks create a macroeconomic trade-off. They need not create steady-state conflict. In the Australian calibration, the no-shock steady state is common to all regimes, but demand and cost-push shocks generate large welfare losses under simultaneous non-cooperation. The cost-push case provides the strongest example of strategic complementarity in instrument levels: fiscal expansion and monetary tightening encourage one another while offsetting in their effect on demand.

The existing open-loop Julia model is therefore informative but preliminary. It identifies the policy-mix externality and supplies a reproducible benchmark. The original draft's emphasis on prior policy, adjustment costs, shocks, and endogenous commitment calls for a stochastic Markov-perfect equilibrium, a feedback version of leadership, and a separate ex ante institutional game. Those extensions are not cosmetic. They determine whether the apparent leadership equivalence survives reoptimisation and whether strategic complementarity creates multiple, unstable, or socially inferior coordination outcomes in the states where it actually matters.

A Reproducibility

All calculations and figures are produced by the Julia project accompanying this paper. From the project directory, run

```
julia --project=. -e "using Pkg; Pkg.instantiate(; update_registry=false)"
julia --project=. run_model.jl
julia --project=. test/runtests.jl
julia --project=. build_tex.jl
```

The first command installs exact package versions from the manifest. The model runner writes calibration primitives, summary tables, the zero-shock audit, the shock-scaling experiment, and all figures to `output/`. The final command invokes the Tectonic binary supplied by Julia's artifact system and creates `PAPER.pdf` from this native $\text{\LaTeX}$ source.

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